

Assignment 4 COMP 551

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Abstract

In this project, we investigated the performance of `bert-base-uncased` on the AG News classification task using two strategies: probing with frozen representations and full end-to-end fine-tuning. The AG News dataset consists of 120,000 news headlines and descriptions across four balanced categories, making it a strong benchmark for document classification.

In the probing setup, we extracted sentence embeddings from a frozen BERT model using four strategies: [CLS] token, first token, last token, and mean pooling. These embeddings were then used as features for lightweight classifiers: K-Nearest Neighbors (KNN) and logistic regression. Mean pooling consistently outperformed other embedding methods, achieving 91.88% test accuracy with KNN and 91.26% with logistic regression, indicating the strength of BERT’s pretrained representations even without finetuning.

We then fine-tuned the BERT model on a reduced training set of 20,000 examples and achieved a test accuracy of 93.12%. This highlights that while pretrained features are already powerful, end-to-end optimization yields additional performance gains by adapting to task-specific patterns.

Lastly, we visualized attention matrices to interpret model behavior. Correct predictions often showed attention concentrated on domain-specific keywords, while misclassifications arose from ambiguous or overlapping vocabulary. Together, these findings highlight the complementary strengths and interpretability trade-offs of probing and fine-tuning.

1 Introduction

Transformer-based models like BERT have redefined the state of the art in natural language processing by enabling robust generalization across a wide range of downstream tasks [1]. Pre-trained on large collections of documents using self-supervised objectives, these models capture rich contextual representations that can be leveraged with minimal task-specific supervision [2]. However, the practical effectiveness of such representations often depends on the fine-tuning strategy and how information is extracted from intermediate model states [3].

In this assignment, we investigated two contrasting approaches to using BERT on the AG News classification task, a four-class dataset covering World, Sports, Business, and Sci/Tech categories. We evaluated *probing* with frozen embeddings and *fine-tuning* the model end-to-end. The probing setup allows us to examine how well pretrained representations serve out-of-the-box for classification when used as input to simple classifiers like K-Nearest Neighbors (KNN) and logistic regression. In contrast, fine-tuning adjusts the model’s internal parameters using supervised labels, enabling it to learn task-specific representations at the cost of increased computation. While effective, full fine-tuning can be parameter-inefficient in multi-task settings, motivating alternatives such as adapter modules [4] that allow for lightweight task-specific adaptation without modifying the entire model.

We systematically compare four sentence embedding strategies from frozen BERT: [CLS] token, first token, last token, and mean pooling, across both classifiers, evaluating their relative performance. In parallel, we fine-tune the full model on a reduced subset of the training data to assess performance gains when internal representations are allowed to adapt.

Finally, we complement our quantitative results with attention matrix visualizations that provide qualitative insights into the model’s focus during classification. This dual approach, empirical and interpretive, allows us to assess both the performance and internal reasoning of BERT-based models in practical text classification settings.

2 Datasets

For this assignment, we used the AG News dataset, which is a collection of news articles labeled into four categories: World, Sport, Business, and Sci/Tech. Each example in the dataset includes a news headline and a short description. The full training set contains 120,000 samples, and the test set contains 7,600.

We accessed the dataset using the `datasets` library from Hugging Face, which made it easy to load and work with. To evaluate different methods fairly, we split the original training data into 80% for training and 20% for validation.

To prepare the data for BERT, we used the `bert-base-uncased` tokenizer. This tokenizer lowercases all text and ensures each input is tokenized to the same length by truncating or padding to a maximum of 128 tokens.

We confirmed that the class distribution is balanced and reviewed sample entries to understand the data structure. Overall, the dataset was clean and ready to use without any preprocessing.

3 Benchmark

We used the pre-trained `bert-base-uncased` model, which has ~ 110 million parameters. It includes 12 transformer layers and 12 attention heads. We tested the model in two different setups: *probing* and *fine-tuning*.

3.1 Probing (Frozen BERT)

In the probing setup, we used BERT as a feature extractor and froze all its internal parameters. We tested four different strategies to generate sentence embeddings:

- **[CLS]**: Used the embedding of the [CLS] token from the final hidden layer.
- **First**: Used the first token embedding after [CLS].
- **Last**: Used the token embedding before [SEP] token, determined from the attention mask.
- **Mean**: Averaged all token embeddings, ignoring any padding.

These sentence embeddings were then used as input features for two classifiers:

- **K-Nearest Neighbors (KNN)**: We tested $K = 1$ to 20 and selected the optimal value using validation accuracy.
- **Logistic Regression**: Trained on the same embeddings, with a capped 1000 iterations. This setup aligns with prior work that uses frozen BERT features as general-purpose sentence representations without fine-tuning [2].

We selected [CLS], first, last, and mean pooling based on prior empirical studies comparing sentence-level pooling methods [3].

3.2 Fine-tuning BERT

For the second setup, we fine-tuned all parameters of BERT by adding a classification head on top. Fine-tuning allows the model to adapt its internal representations to the specifics of the AG News classification task, in contrast to probing which uses fixed embeddings. The model was trained on a reduced subset of 20,000 training and 2,000 validation examples to remain within modest compute limits.

We used the `Trainer` class from the Hugging Face `transformers` library to manage training and evaluation. Training was performed for up to 3 epochs with early stopping triggered if validation accuracy did not improve after 1 epoch. A linear learning rate scheduler was applied without warmup steps. Accuracy was used as the evaluation metric for both model selection and performance comparison. A summary of the fine-tuning hyperparameters is provided in Table 1.

Table 1: Summary of Fine-Tuning Hyperparameters

Parameter	Value
Learning Rate	2×10^{-5}
Batch Size	16
Optimizer	AdamW
Weight Decay	0.01

To address the parameter inefficiency of full fine-tuning in multi-task settings, recent work has proposed lightweight alternatives such as adapter modules [4]. These methods train only a small number of additional parameters per task, making them especially promising for scaling BERT-like models to multiple domains without retraining the entire model.

4 Results

4.1 Probing

We evaluated a frozen BERT model’s ability to capture semantic information for classification by passing a subset of the training data through it to produce sentence-level embeddings. Specifically, we considered four embedding strategies: utilizing the [CLS] token representation from the final hidden layer, using the first or last token embedding, and computing the mean of all token embeddings. After extracting embeddings, we used them as input features for multi-class Logistic Regression and K-nearest neighbors (KNN). We optimized K on a validation split.

Results in Table 2 indicate that different embedding strategies yield varying performance. The mean embedding outperformed other strategies (86.70% with logistic regression, 88.90% with KNN at K=10), while the first token embedding underperformed. This suggests that averaging embeddings captures richer semantic information than relying on a single token’s embedding, consistent with prior findings favoring mean pooling for general sentence-level tasks [3].

Table 2: Validation Accuracy by Embedding Strategy and Classifier

Embedding Strategy	Logistic Regression Accuracy	KNN Best Accuracy	Best K
cls	85.20%	83.30%	20
first	77.60%	70.60%	8
last	82.00%	77.40%	5
mean	86.70%	88.90%	10

For the KNN specifically, Figure 1 provides additional insight by visualizing the validation accuracy trends across varying neighbors (K) for each embedding strategy. The mean embedding consistently outperforms the others, indicating its robustness. The [CLS] strategy is relatively stable, improving slightly with larger K , while the last token embedding is moderate but still weaker. The first token embedding performs the worst, reflecting its limited semantic capture.

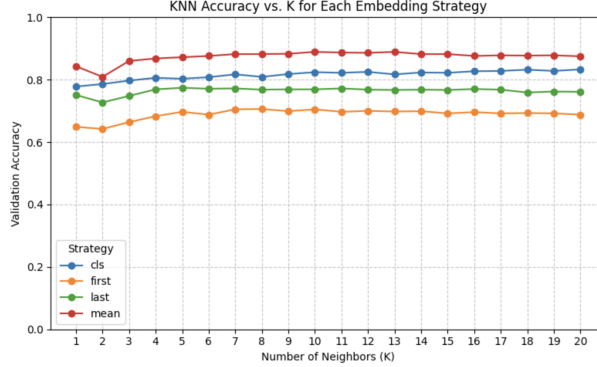


Figure 1: Validation accuracies for different K and Embedding strategy

4.2 Fine-tuning

Following our probing experiments, we proceeded to fine-tune all parameters of the pretrained BERT model specifically on the AG News classification task. Unlike the probing stage, fine-tuning involves updating the internal weights of the transformer architecture through stochastic gradient descent (SGD). Due to computational constraints, we fine-tuned BERT on a reduced dataset selecting 20,000 examples for the training.

We fine-tuned the BERT model for 3 epochs, using a validation subset to monitor training progress and prevent overfitting. Training was halted when validation accuracy improvements plateaued or began to decline. The mean embedding strategy, identified as the best-performing embedding strategy from the probing experiments, served as a comparative baseline.

Table 3 summarizes the training and validation metrics observed across epochs during fine-tuning. We achieved optimal validation accuracy of 92.55% by the third epoch, with the loss beginning to show an increase afterward, indicating potential overfitting.

Table 3: Training and Validation Metrics across Epochs

Epoch	Training Loss	Validation Loss	Validation Accuracy
1	0.316400	0.289575	91.05%
2	0.174600	0.279180	92.35%
3	0.106000	0.338194	92.55%

The fine-tuned BERT achieved higher validation accuracy than all previous methods used during probing. This analysis underscores the substantial performance gains achievable by fine-tuning BERT’s parameters, validating its effectiveness in adapting pretrained transformers for specific classification tasks.

4.3 Classification performances

We evaluated the multiclass classification **test accuracy** on the AG News test dataset using the best-performing probing strategies—K-nearest neighbors (KNN) and logistic regression

(LogReg)—as well as the end-to-end fine-tuned BERT model. Both KNN and logistic regression utilized the mean embedding strategy, with the optimal KNN performance using $K = 10$ (corresponding to the best K found previously). Figure 2 presents these test accuracies.

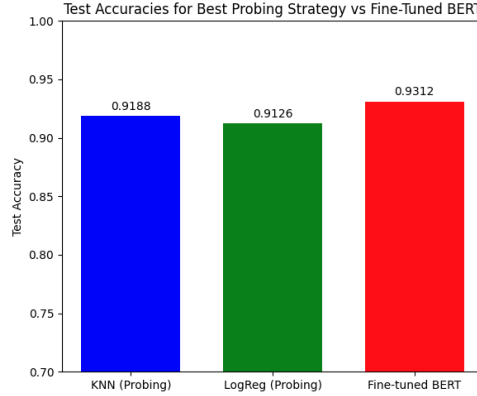


Figure 2: Test Accuracies for Best Probing Strategies vs Fine-Tuned BERT

As shown in Figure 2, we observe that the fine-tuned BERT model achieves the highest accuracy (93.12%), outperforming both probing approaches (91.88% for KNN and 91.26% for logistic regression). The improvement from probing to fine-tuning is notable, highlighting the effectiveness of end-to-end training.

This performance gain can be attributed to the comprehensive updating of all transformer weights during fine-tuning, allowing the model to capture the unique patterns of the AG News dataset. In contrast, probing approaches leverage fixed embeddings from a pretrained model. Thus, while probing provides insight into BERT’s capabilities, fine-tuning significantly enhances flexibility and performance by enabling the model to adjust fully to the task-specific context.

4.4 Attention Matrix Visualization

To further examine how our fine-tuned BERT model makes predictions, we visualized the attention weights in the final (12th) transformer layer. Specifically, we looked into an attention head to see which tokens [CLS] focuses on the most. Although each attention head can learn unique patterns, viewing even one head provides insight into the model’s reasoning [5]. However, it is important to note that attention weights alone may not provide a complete explanation of model behavior [6]. We show **correctly** and **incorrectly** predicted examples for each label (0–3).

Furthermore, while some works argue that attention can offer valuable insight into model reasoning [5], others emphasize that attention alone is not necessarily faithful as an explanation [7]. As such, attention heatmaps should be treated as heuristic visual aids rather than definitive interpretive tools.

Document Snippets

In this section, we present full sentences for two AG News labels—World (0) and Sports (1). For each, we include one correctly predicted snippet and one misclassified. Although BERT was trained on four categories, focusing on these two illustrates how overlapping vocabulary can cause confusion. For instance, domain-specific terms tend to lead to correct predictions, while ambiguous language or overlapping terms can cause misclassifications. The examples below reflect this pattern, though the exact reasoning remains uncertain due to the model’s complexity.

- **Label 0 (World)**

- **Correctly Predicted:**

*Sister of man who died in Vancouver police custody slams chief (Canadian Press)
Canadian Press - VANCOUVER (CP) - The sister of a man who died after a violent confrontation with police has demanded the city's chief constable resign for defending the officer involved.*

- **Incorrectly Predicted:**

Olympics day four Richard Faulds and Stephen Parry are going for gold for Great Britain on day four in Athens.

- **Label 1 (Sports)**

- **Correctly Predicted:**

Giddy Phelps Touches Gold for First Time Michael Phelps won the gold medal in the 400 individual medley and set a world record in a time of 4 minutes 8.26 seconds.

- **Incorrectly Predicted:**

Glory Comes Amid Empty Seats and Closed Shutters HERE in Old Europe, people install shutters outside their windows to keep out the heat, the pollution, the daylight, the noise. They also lock the shutters tight when they go away on holiday.

Interestingly, the snippet from Label 0 (World) mentioning “Olympics” and “gold” was misclassified as Sports, while the Sports snippet referencing “Europe” and “holiday” was predicted as World. This mutual confusion underscores how a few overlapping words can overshadow the article’s true theme, misleading the model to swap categories.

While we focused our analysis on a single head in the final transformer layer, prior work suggests that aggregating attention across heads or layers—such as via mean pooling or max-attention heuristics—can provide more stable insights [6].

Attention Matrix

Here, we visualize how the [CLS] token attends to the text for the same examples shown above. Each heatmap corresponds to one of the final transformer’s heads (the first head in the 12th layer). In Transformer models like BERT, each attention head can learn unique semantic relationships and patterns in the text. By examining these patterns for both correctly and incorrectly predicted examples, we identify which tokens the model deems most indicative of a given AG News label.

Label 0 (World)

Here, the [CLS] token often attends to domain-specific words (e.g., “olympics,” “gold,” “Vancouver,” “Canadian”). However, in cases with overlapping terms like “Olympic,” attention may split across domains, leading to misclassification. The heatmaps highlight these partial, head-specific terms that shape the model’s final prediction.

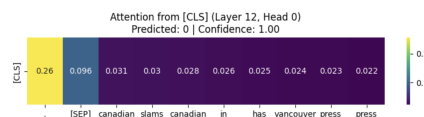


Figure 3: Attention heatmap for a *World* article correctly predicted as Label 0.

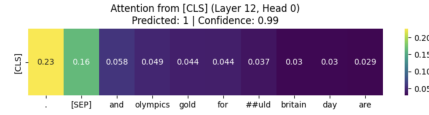


Figure 4: Attention heatmap for a *World* article misclassified by the model.

Label 1 (Sports)

For sports examples, this head tends to focus on performance-related terms like “record” and athletes when making correct predictions. However, the incorrect snippet featuring terms like “Europe” and “holiday” was misclassified as World (Label 0), likely because these overlapping cues pulled the model toward the wrong category.

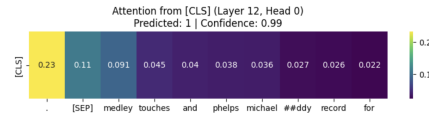


Figure 5: Attention heatmap for a *Sports* article correctly predicted as Label 1.

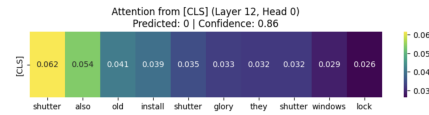


Figure 6: Attention heatmap for a *Sports* article misclassified by the model.

Label 2 (Business)

In this scenario, the head focuses on financial terms like “pension” and “firm” when predicting Business. Tech-related words like “chip” or “TV” also attract attention, showing overlap between domains. In this case, tech references led to a misclassification as Sci/Tech (Label 3).

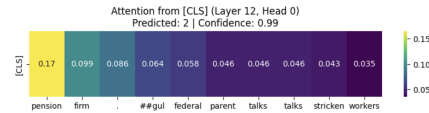


Figure 7: Attention heatmap for a *Business* article correctly predicted as Label 2.

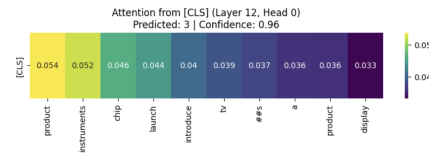


Figure 8: Attention heatmap for a *Business* article misclassified by the model.

Label 3 (Sci/Tech)

In this case, this head focused on scientific terms like “rocket” and “space” when identifying Sci/Tech content. However, in the incorrect prediction, financial terms were more dominant than technical ones potentially leading to a misclassification as Business (Label 2).

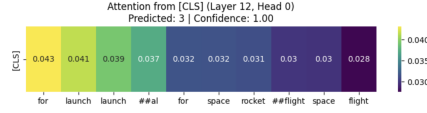


Figure 9: Attention heatmap for a *Sci/Tech* article correctly predicted as Label 3.

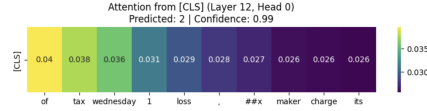


Figure 10: Attention heatmap for a *Sci/Tech* article misclassified by the model.

It is apparent that when a snippet contains highly indicative or domain-specific words, those words tend to receive stronger attention. Conversely, snippets with more ambiguous or overlapping vocabulary can show a more diffuse or inconsistent attention distribution.

Overall, these heatmaps suggest that the model’s attention is unevenly distributed across tokens, with the [CLS] token often focusing on a small subset of semantically important words. In some cases, attention is sharply concentrated on a few tokens, while in others it is more evenly spread across several terms. This variation may reflect the model’s effort to capture different aspects of the text it considers most relevant. It’s important to note that these attention visualizations reflect only one of many heads in the BERT model. While they highlight patterns the selected head focuses on, the final prediction results from all layers and heads combined. Thus, focusing on a single head’s perspective may be somewhat misleading if we attempt to draw firm conclusions about how the entire network reasons. Instead, these heatmaps offer a partial view into BERT’s reasoning, not a complete explanation.

5 Conclusion and Future Work

This project compared two paradigms for leveraging `bert-base-uncased` on the AG News classification task: probing with frozen embeddings and full fine-tuning. While probing demonstrated that BERT’s pretrained features, particularly those derived from mean pooling, are highly informative for lightweight classifiers, end-to-end fine-tuning achieved the best overall performance at 93.12% accuracy, showing the added benefit of task-specific adaptation.

Attention visualizations provided interpretability insights, revealing that [CLS] often focuses on meaningful keywords during correct predictions but can be misled by ambiguous terms in edge cases. These findings highlight both the representational richness and decision complexity inherent in transformer models, as observed in prior studies examining BERT’s internal attention dynamics [5]. While probing offers an efficient baseline with interpretable outputs [2], fine-tuning unlocks full model capacity.

Future work could also explore more robust interpretability frameworks by averaging or contextualizing attention maps across heads and layers [6], and evaluate model robustness using adversarial inputs that subtly alter input meaning while misleading predictions [8]. Additionally, exploring alternative pretraining paradigms like the T5 text-to-text framework [9] may offer a unified architecture for handling diverse NLP tasks under a consistent interface.

6 Statement of Contribution

Leen Assaf: Contributed to the design and implementation of Tasks 1, 2, and 3, including probing experiments and fine-tuning procedures, and drafting the final report.

Karl Wehbe: Worked on the execution and evaluation of Tasks 1, 2, and 3, helped fine-tune BERT, generate attention visualizations, and drafting the report.

Shyam Desai: Involved in developing and testing all three tasks, focusing on embedding strategies and probing performance, and drafting the final report.

References

- [1] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019.
- [2] Harish Tayyar Madabushi, Elena Kochkina, and Michael Castelle. Cost-sensitive bert for generalisable sentence classification with imbalanced data, 2020.
- [3] Jinming Xing, Dongwen Luo, Chang Xue, and Ruilin Xing. Comparative analysis of pooling mechanisms in llms: A sentiment analysis perspective, 2025.
- [4] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin De Larous-silhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2790–2799. PMLR, 09–15 Jun 2019.
- [5] Kevin Clark, Urvashi Khandelwal, Omer Levy, and Christopher D. Manning. What does bert look at? an analysis of bert’s attention, 2019.
- [6] Kuncahyo Setyo Nugroho, Anantha Yullian Sukmadewa, and Novanto Yudistira. Large-scale news classification using bert language model: Spark nlp approach. In *6th International Conference on Sustainable Information Engineering and Technology 2021*, SIET ’21, page 240–246. ACM, September 2021.
- [7] Sarah Wiegrefe and Yuval Pinter. Attention is not not explanation, 2019.
- [8] Robin Jia and Percy Liang. Adversarial examples for evaluating reading comprehension systems, 2017.
- [9] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.